

CSA1701-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

2. 8-Queen Problem

Aim:

To develop a Python program that solves the 8-Queen Problem using backtracking to place 8 non-attacking queens on an 8x8 chessboard.

Algorithm:

1. Represent the chessboard as an 8x8 matrix initialized to 0.
2. Start placing queens in the leftmost column.
3. For the current column, check each row. If placing a queen is safe (no other queens in the same row or diagonals), place it.
4. Recursively attempt to place queens in the remaining columns.
5. If placing the queen leads to a solution, return true.
6. If it does not lead to a solution, backtrack (remove the queen) and try the next row.
7. Print the board if a valid arrangement is found.

```

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def print_solution(board):
    for row in board:
        print(" ".join("Q" if c else "." for c in row))
    print()

def is_safe(board, row, col):
    for i in range(col):
        if board[row][i]: return False
    for i, j in zip(range(row, -1, -1), range(col, -1, -1)):
        if board[i][j]: return False
    for i, j in zip(range(row, 8, 1), range(col, -1, -1)):
        if board[i][j]: return False
    return True

def solve_n_queens(board, col):
    if col >= 8:
        return True
    for i in range(8):
        if is_safe(board, i, col):
            board[i][col] = 1
            if solve_n_queens(board, col + 1):
                return True
            board[i][col] = 0 # Backtrack
    return False

board = [[0] * 8 for _ in range(8)]
if solve_n_queens(board, 0):
    print("Solution Found for 8-Queens:")
    print_solution(board)
else:
    print("No solution exists.")

```

Output:

```

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Python 3.11.9 (tags/v3.11.9:de54cf5, Apr 2 2024, 10:12:12) [MSC v.1938 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/VISHAL M/Downloads/ai expl.py
Solution Found for 8-Queens:
Q . . . . .
. . . . . Q .
. . . . Q . .
. . . . . . Q
. Q . . . . .
. . . Q . . .
. . . . . Q .
. . Q . . . .

```

Result:

Thus, the 8-Queen Problem solved successfully using Python.